Heart Disease Prediction using Machine Learning

A ML Project

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Project Objectives

Data Analysis & Risk Identification

To analyze patient medical data and identify key clinical risk factors that contribute to the presence of heart disease.

Model Development & Comparison

To build and evaluate multiple machine learning models including Logistic Regression, Random Forest, and SVM to find the most effective predictor.

Performance Optimization

To optimize model performance with a focus on high recall, ensuring that potential heart disease cases are not missed by the system.

Interactive Deployment

To deploy the final model as a user-friendly Streamlit dashboard for real-time clinical risk assessment.

CORE GOALS

Problem Statement

Global Health Crisis

Cardiovascular diseases (CVDs) are the leading cause of death globally, taking an estimated 17.9 million lives each year.

Critical Need for Early Detection

Early identification of high-risk individuals is essential for effective treatment and preventing premature deaths.

Diagnostic Challenges

Manual diagnosis based on clinical parameters can be time-consuming and prone to human error, especially in resource-limited settings.

The ML Solution

An automated, data-driven system can provide rapid, accurate screening to assist medical professionals in early diagnosis.

17.9M

ANNUAL DEATHS GLOBALLY

32%

OF ALL GLOBAL DEATHS

Dataset Overview

Feature Name	Description	Type
Age / Sex	Patient age and gender (1=Male, 0=Female)	Numeric / Binary
Chest Pain (cp)	Type of chest pain experienced (Value 0-3)	Categorical
Trestbps	Resting blood pressure (mm Hg on admission)	Numeric
Chol	Serum cholestoral in mg/dl	Numeric
Thalach	Maximum heart rate achieved	Numeric
Target	Diagnosis of heart disease (1=Yes, 0=No)	Binary (Label)

DATASET SOURCE

UCI Cleveland

TOTAL SAMPLES

303 Rows

TOTAL FEATURES

14 Attributes

UCI Machine Learning Repository

Data Preprocessing

Missing Value Management

The dataset was audited for null values. While the Cleveland subset is largely clean, a pipeline was established to handle any potential missing data using mean imputation for numerical features.

Feature Scaling (Standardization)

Applied **StandardScaler** to normalize features like Age, Cholesterol, and Blood Pressure. This ensures that features with larger magnitudes do not dominate the model's learning process.

Categorical Encoding

Categorical variables such as Sex, Chest Pain Type, and Thalassemia were verified for proper numerical representation to ensure compatibility with Scikit-Learn algorithms.

Stratified Data Splitting

Data was split into training and testing sets using stratification. This maintains the same proportion of heart disease cases in both sets, preventing bias during evaluation.

80:20

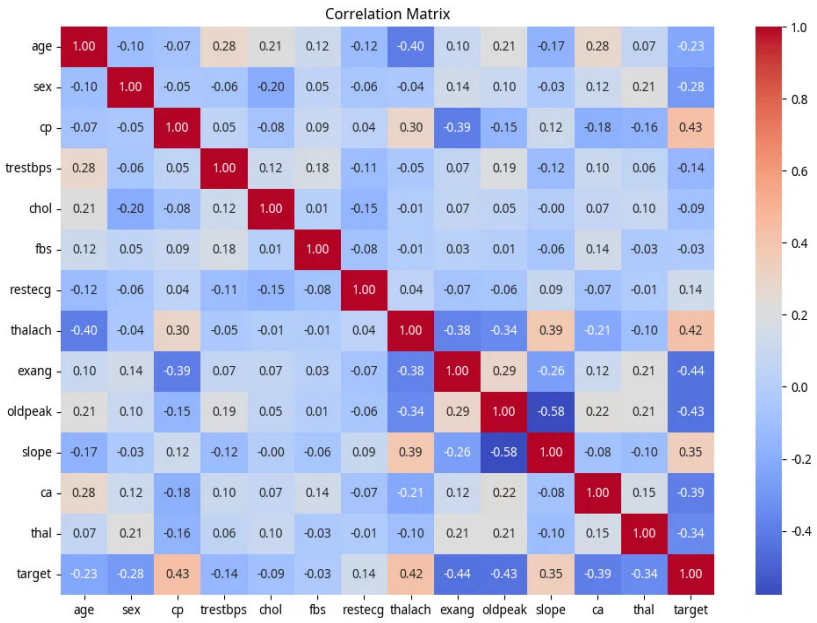
TRAIN-TEST SPLIT

303

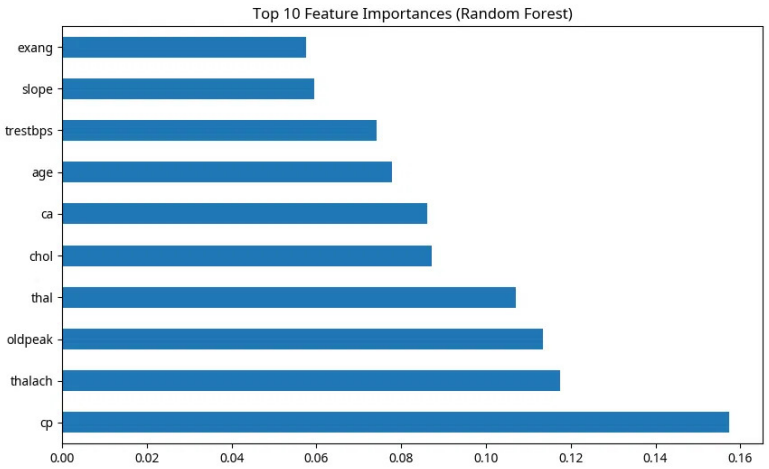
TOTAL SAMPLES

Exploratory Data Analysis (EDA)

Correlation Heatmap



Top Feature Importance



Key Insights

Strong Correlations

Chest pain type (cp) and maximum heart rate (thalach) show the strongest positive correlation with heart disease.

Risk Factors

Exercise-induced angina (exang) and ST depression (oldpeak) are significant negative predictors.

Clinical Significance

The number of major vessels (ca) is a critical feature, as confirmed by the Random Forest importance plot.

Machine Learning Algorithms

Logistic Regression

Interpretability: Serves as a strong baseline model. It provides clear insights into how each clinical feature affects the probability of heart disease.

Random Forest

Robustness: An ensemble method that handles non-linear relationships and reduces overfitting by averaging multiple decision trees.

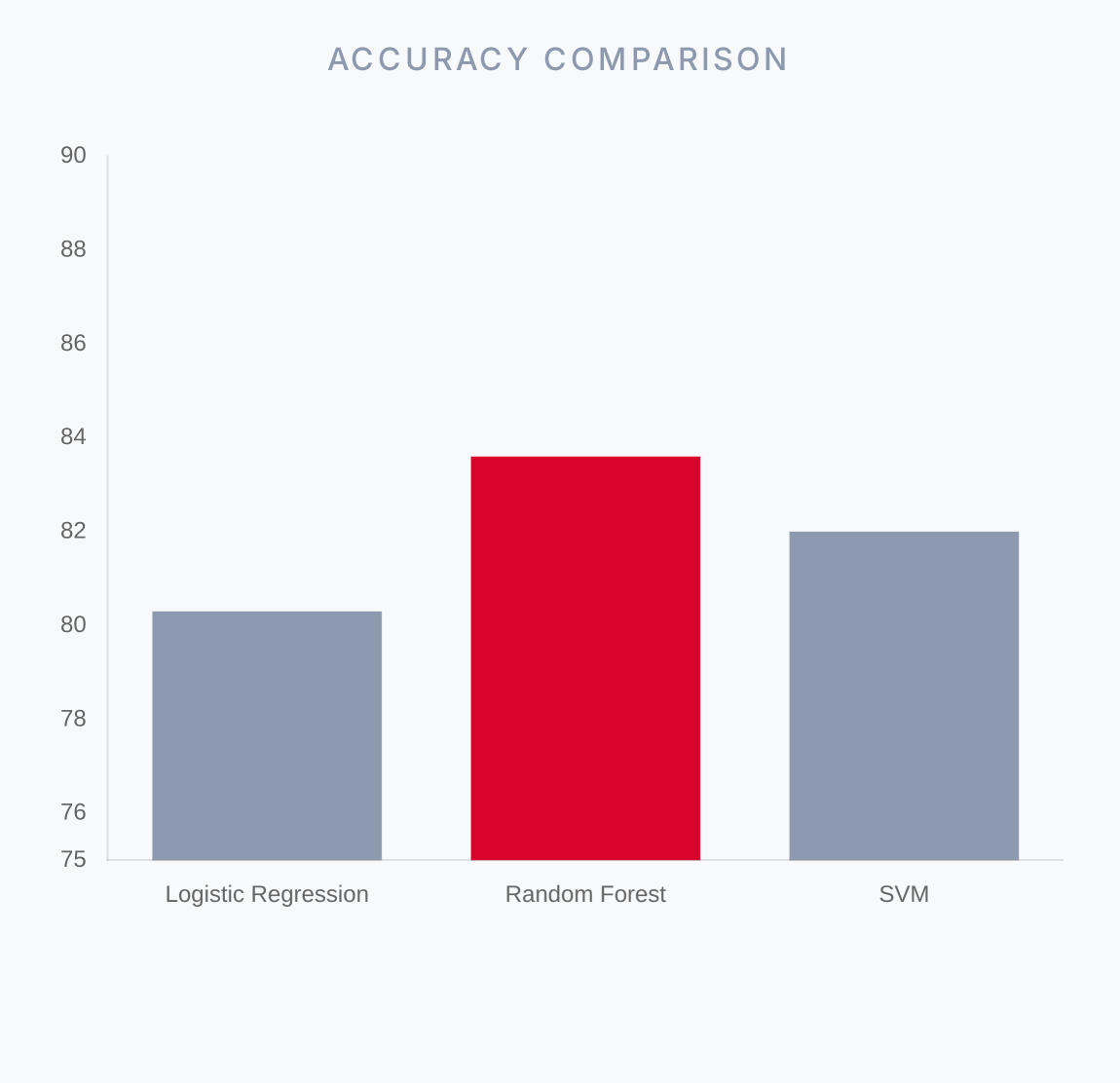
Support Vector Machine

Efficiency: Highly effective in high-dimensional spaces. It works by finding the optimal hyperplane that maximizes the margin between classes.

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	80.3%	76.9%	90.9%	83.3%
Random Forest	**83.6%**	**78.0%**	**97.0%**	**86.5%**
SVM	82.0%	77.5%	93.9%	84.9%

* Metrics evaluated on a 20% hold-out test set.



Final Model Selection

Superior Accuracy & Generalization

The Random Forest model achieved the highest overall accuracy of 83.6%, demonstrating its ability to generalize well on unseen patient data compared to Logistic Regression and SVM.

High Recall (Sensitivity)

With a recall of 97.0%, the model is exceptionally good at identifying actual heart disease cases. In medical screening, minimizing "False Negatives" is the highest priority.

Robustness to Non-Linearity

As an ensemble method, Random Forest effectively captures complex, non-linear relationships between clinical features like Age, Cholesterol, and Blood Pressure without overfitting.

Feature Importance Insights

The model provides built-in feature importance scores, allowing clinicians to understand which factors (like chest pain or major vessels) are driving the risk assessment.

Random Forest

97.0%
RECALL RATE

83.6%
ACCURACY

Results & Insights

High Sensitivity (Recall)

The Random Forest model achieved a **97.0% Recall**, which is critical in medical diagnostics. This means the system correctly identifies almost all patients with heart disease, minimizing dangerous false negatives.

Clinical Feature Significance

Analysis confirms that **Chest Pain Type (cp)** and **Number of Major Vessels (ca)** are the most influential predictors, aligning with established medical knowledge of cardiovascular risk factors.

Model Reliability

With an overall **Accuracy of 83.6%** and a balanced F1-Score, the model demonstrates robust performance across both healthy and diseased patient profiles in the test dataset.

KEY METRIC

97% Recall Rate

TOP PREDICTOR

Chest Pain Type

CLINICAL IMPACT

Early Screening Support

Streamlit Dashboard Preview

Interactive Sidebar Inputs

Users can input clinical parameters like Age, Blood Pressure, and Cholesterol using intuitive sliders and dropdown menus.

Real-time Prediction Engine

The backend processes inputs through the trained Random Forest model to provide instant risk assessments (High vs. Low Risk).

Probability Visualization

Displays the confidence level of the prediction, helping medical professionals understand the certainty of the model's output.

Clinical Decision Support

Designed as a secondary screening tool to assist doctors in prioritizing patients for further diagnostic testing.

TECHNOLOGY STACK

Streamlit

Python

Scikit-Learn

Pandas

Live Demo Available

Conclusion & Future Scope

Conclusion

Effective Prediction

Machine learning models, particularly Random Forest, demonstrate high efficacy in identifying heart disease risk from clinical data.

Clinical Utility

The high recall rate (97%) ensures that the system is a reliable screening tool for medical professionals to minimize false negatives.

Accessible Technology

The Streamlit dashboard provides an intuitive interface for real-time risk assessment without requiring technical expertise.

Future Scope

IoT Integration

Connecting the model to real-time wearable health monitors for continuous patient tracking and early warning alerts.

Advanced Architectures

Exploring Deep Learning and Neural Networks to potentially capture even more complex patterns in larger datasets.

Broader Data Features

Incorporating lifestyle factors, genetic data, and dietary habits to improve the model's holistic predictive power.

Thank You .